

DHAHARAN KING ABDULAZIZ AB, Saudi ArabiaWMO: **404160**Lat: **26.265N**Lon: **50.152E**Elev: **26**StdP: **101.02**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.1	9.5	-4.2	2.7	18.5	-2.0	3.2	20.8	10.5	16.9	9.8	16.9	4.3	300	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	45.9	22.9	44.8	22.9	43.2	23.0	31.3	35.9	30.4	35.6	29.4	35.2	5.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.2	27.6	34.3	29.1	25.9	33.9	28.0	24.3	33.8	108.1	35.4	102.5	35.2	97.9	35.0	36.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.7	8.8	5.6	47.9	1.6	1.2	4.5	48.7	3.6	49.4	2.7	50.1	1.5	51.0		
			2.0	32.7	1.9	1.5	0.6	33.8	-0.5	34.7	-1.6	35.5	-3.0	36.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	16.4	18.1	21.9	27.0	32.7	35.8	37.1	36.6	33.8	29.8	23.4	18.2
	DBStd	7.82	2.60	2.89	2.95	2.93	2.42	1.78	1.39	1.47	1.95	2.20	3.06	2.79
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	151	72	36	4	0	0	0	0	0	0	0	2	37
	CDD10.0	6429	198	226	368	509	705	774	839	825	714	614	402	255
	CDD18.3	3539	11	29	114	259	447	524	581	567	464	356	154	33
	CDH23.3	52418	36	134	864	2879	6882	8903	9986	9548	7220	4572	1250	145
	CDH26.7	35284	2	20	298	1428	4547	6512	7507	7068	4885	2578	429	12
Wind	WSAvg	4.3	4.2	4.6	4.6	4.5	4.6	5.1	4.6	3.8	3.8	3.5	4.0	4.1
Precipitation	PrecAvg	77	15	13	19	7	1	0	0	0	0	1	10	13
	PrecMax	327	82	78	208	59	11	0	0	0	0	18	121	94
	PrecMin	5	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	66	18	20	35	10	2	0	0	0	0	3	22	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	29.1	35.9	40.0	45.0	47.1	47.2	46.9	44.8	41.0	35.0	28.1
		MCWB	15.8	17.2	18.2	20.3	21.9	21.8	23.2	23.3	22.9	21.3	21.2	18.8
	2%	DB	24.1	26.8	32.0	37.2	42.8	45.2	45.9	45.0	42.9	39.0	32.2	26.2
		MCWB	16.3	16.1	17.4	19.9	21.7	22.1	23.1	24.0	22.9	22.0	21.0	17.8
	5%	DB	22.8	24.9	29.9	35.2	41.0	43.9	44.5	43.9	41.4	37.2	30.9	25.0
		MCWB	16.0	16.0	17.0	19.2	21.1	21.9	22.9	24.3	23.2	22.2	20.8	17.7
	10%	DB	21.2	23.2	27.9	33.2	39.2	42.2	43.2	42.8	40.0	35.9	29.1	23.6
		MCWB	15.3	15.7	16.6	19.2	21.1	21.7	23.0	24.6	23.7	22.3	20.4	17.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	20.3	21.9	24.1	28.2	29.2	32.2	32.8	31.6	28.6	26.2	21.8
		MCDB	22.2	24.2	27.3	31.1	34.8	34.9	35.9	36.2	35.9	32.2	29.5	24.8
	2%	WB	18.5	19.0	20.6	22.7	26.2	27.6	30.7	31.8	30.0	27.4	24.2	20.5
		MCDB	21.4	22.3	25.7	30.1	33.9	34.2	35.9	36.1	34.7	31.2	28.4	23.6
	5%	WB	17.5	18.2	19.7	21.9	24.9	26.2	29.6	30.8	28.9	26.6	23.1	19.4
		MCDB	20.7	21.9	24.9	29.7	33.4	34.8	35.5	35.7	34.2	30.8	27.5	22.7
	10%	WB	16.5	17.3	18.7	21.1	23.7	24.9	28.5	30.0	27.9	25.8	22.1	18.3
		MCDB	20.1	21.3	24.5	29.2	33.5	35.5	35.2	35.6	33.6	30.6	26.7	22.0
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.2	11.2	12.0	13.0	13.7	13.2	12.8	13.3	12.7	10.6	10.0
		MCDBR	10.6	11.7	14.4	14.8	15.4	15.9	15.1	14.6	14.7	14.5	12.2	11.1
		MCWBR	5.6	5.0	5.5	5.2	5.7	6.5	6.9	7.3	7.1	6.3	5.7	5.7
	5% WB	MCDBR	8.7	9.0	10.6	12.4	13.4	14.5	12.7	12.0	12.5	12.5	10.7	9.0
		MCWBR	5.7	5.0	5.3	5.3	6.3	7.4	6.9	6.5	6.8	6.3	5.7	5.5
Clear-Sky Solar Irradiance	taub		0.520	0.571	0.632	0.692	0.687	0.656	0.686	0.636	0.565	0.527	0.533	0.501
	taud		1.839	1.731	1.606	1.541	1.539	1.582	1.570	1.670	1.790	1.870	1.856	1.896
	Ebn at Noon		706	705	691	666	669	686	665	696	734	735	689	702
	Edh at Noon		183	219	261	285	286	272	275	247	214	187	178	166
All-Sky Solar Radiation	RadAvg		3.73	4.46	5.35	5.96	6.85	7.40	7.00	6.67	6.15	5.12	3.85	3.52
	RadStd		0.25	0.28	0.38	0.37	0.20	0.21	0.16	0.13	0.12	0.15	0.27	0.25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.47	N/A	-3.14	+1.30	N/A	N/A	N/A	+204	+202
(47) Regional (5 neighbors)		+0.51	N/A	-1.52	+0.70	N/A	N/A	N/A	+196	+191

Nomenclature: See separate page